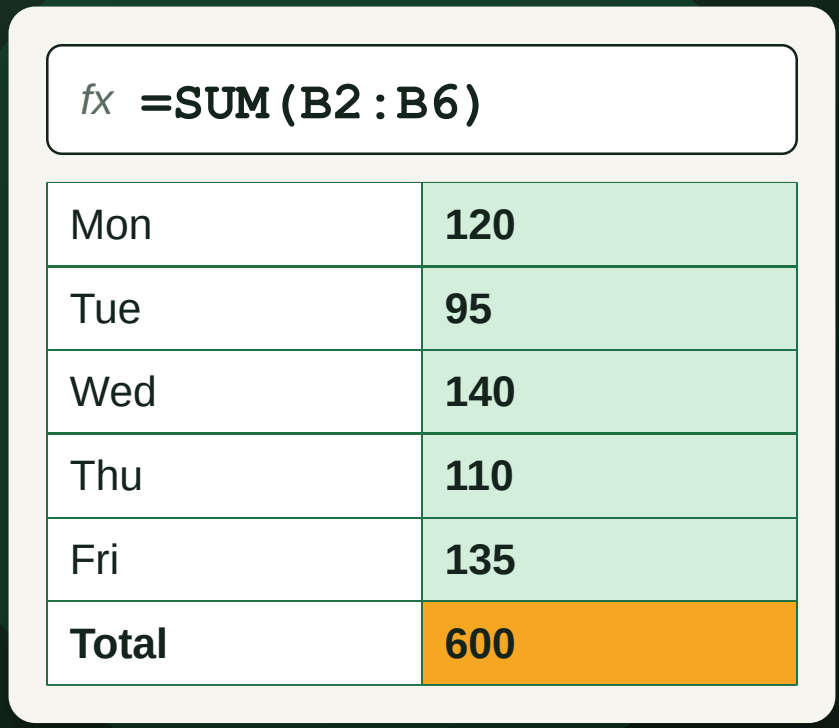
UNIT 8

# Basic Excel Formulas

SUM AVERAGE MIN MAX

Learn it, try it, check it



THEORY · WHAT IT IS

# What is a formula?

A **formula** is an instruction that tells Excel to do a calculation.

1 · YOU TYPE

*fx* =5+3

It starts with =, so Excel knows to calculate

→

2 · THE CELL SHOWS

8

The result appears, not the formula

A formula can use numbers (=5+3) or cells (=B2+B3).

THEORY · WHY IT IS NEEDED

# Why do we need formulas?

BEFORE

| | |
|---|---|
| Mon | **120** |
| Tue | **95** |
| Wed | **140** |
| Thu | **110** |
| Fri | **135** |
| **Total** | **600** |

**Wed changes to 200**

→

AFTER: THE TOTAL UPDATES BY ITSELF

| | |
|---|---|
| Mon | **120** |
| Tue | **95** |
| Wed | **200** |
| Thu | **110** |
| Fri | **135** |
| **Total** | **660** |

**Fast**
one formula, no long sum

**Accurate**
no typing mistakes

**Automatic**
updates by itself

THEORY · KEY TERMS

# Four words to know

THE SHEET

| | A | B |
|---|---|---|
| 1 | **Day** | **Sales** |
| 2 | Mon | **120** |
| 3 | Tue | **95** |
| 4 | Wed | **140** |
| 5 | Thu | **110** |
| 6 | Fri | **135** |

## Cell

One box in the sheet

`B2`

## Range

A group of cells

`B2:B6`

## Function

A ready-made tool

`SUM`

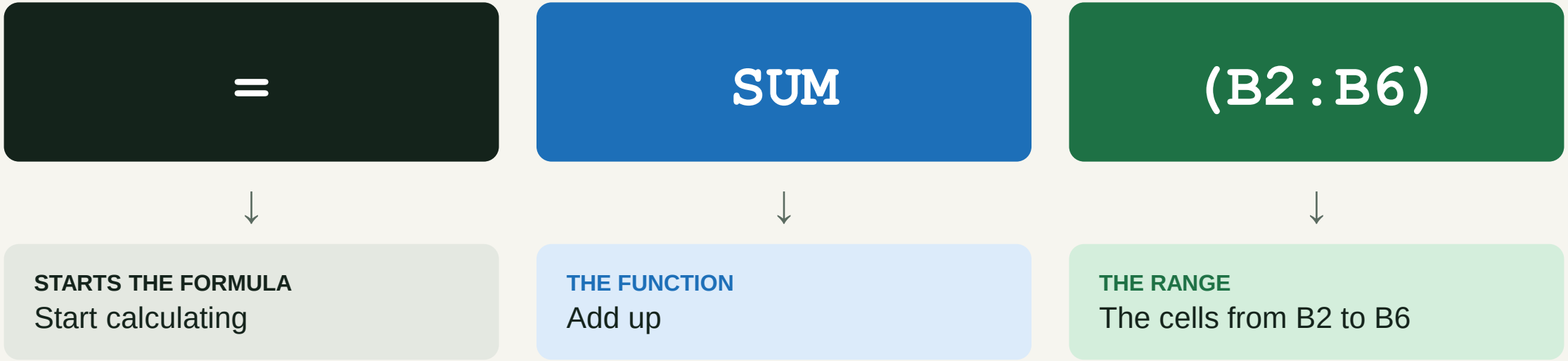
## Formula

Starts with =

`=SUM(B2:B6)`

THEORY · THE PATTERN

# Formula anatomy




A colon means **to**. Read it aloud: "equals, SUM, B2 to B6".

THEORY · HOW TO USE IT

# Entering a formula

**1 Click a cell**

B7

Pick the cell where the answer should go.

**2 Type it**

*fx* `=SUM(B2:B6)`

Start with =, then the function and the range.

**3 Press Enter**

Enter

The cell shows the result.

**Tip:** instead of typing B2:B6, click and drag over the cells.

FUNCTION 1 OF 4

# SUM: add the values

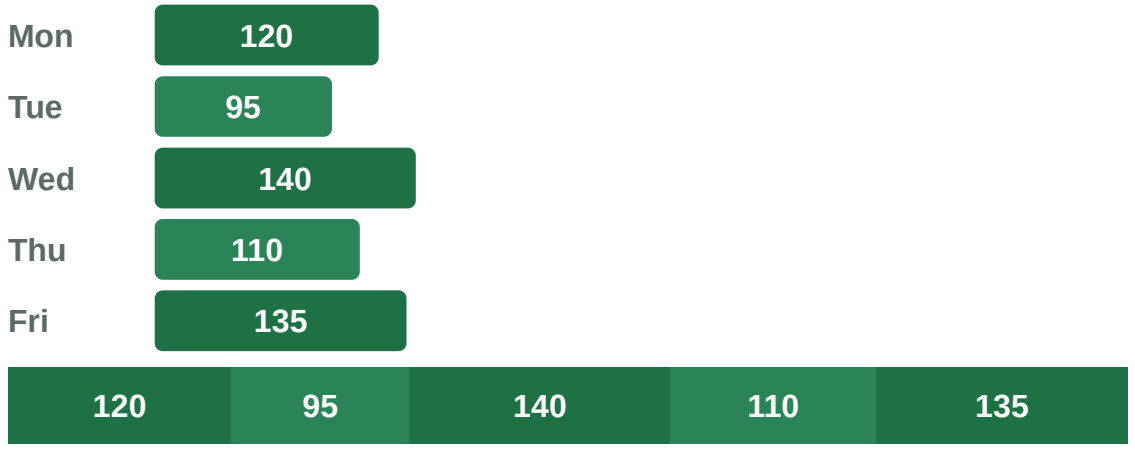
**WHAT IT DOES**

Adds all the numbers in a range.

**EXAMPLE**

`=SUM(B2:B6)` → **600**

**JOIN THE BARS END TO END**




**Total = 600**

**Use it for:** totals, like sales, marks or expenses.

**Watch out:** include every cell, B2:B6 not B2:B5.

FUNCTION 2 OF 4

# AVERAGE: find the typical value

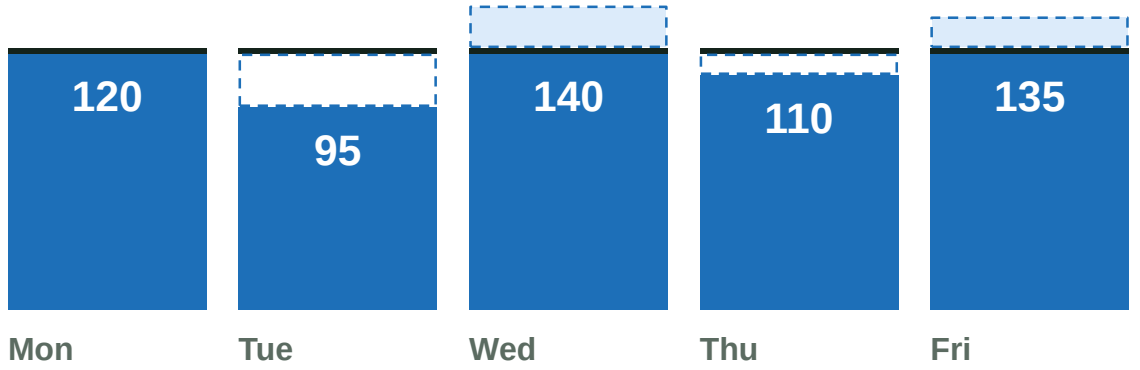
**WHAT IT DOES**

Adds the numbers, then divides by how many there are.

**EXAMPLE**

`=AVERAGE(B2:B6)` → **120**

**THE LINE IS THE AVERAGE: IF EVERY DAY WERE EQUAL, ALL BARS WOULD REACH IT**




**Use it for:** the typical value, like the average mark.

**Watch out:** an empty cell is skipped, but a 0 counts.

FUNCTION 3 OF 4

# MIN: find the smallest value

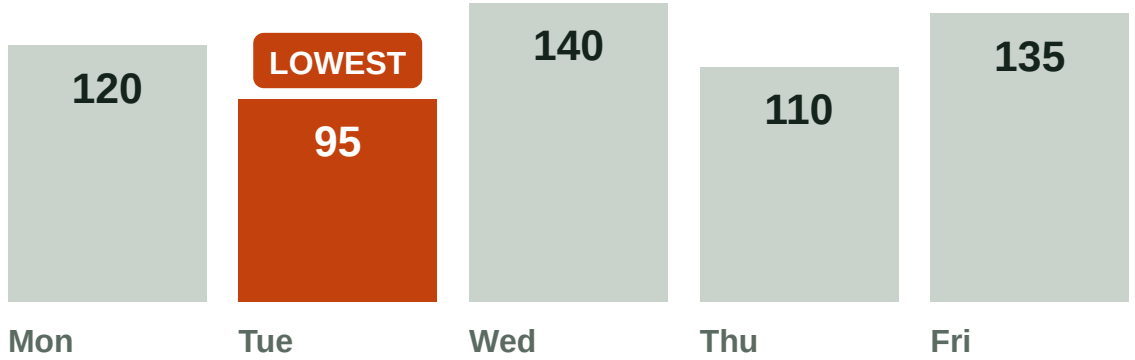
**WHAT IT DOES**

Finds the smallest number in a range.

**EXAMPLE**

`=MIN(B2:B6)` → **95**

**THE SMALLEST BAR IS THE MIN**




**Use it for:** the lowest price, mark or slowest day.

**Watch out:** only numbers count, text in the range is ignored.

FUNCTION 4 OF 4

# MAX: find the largest value

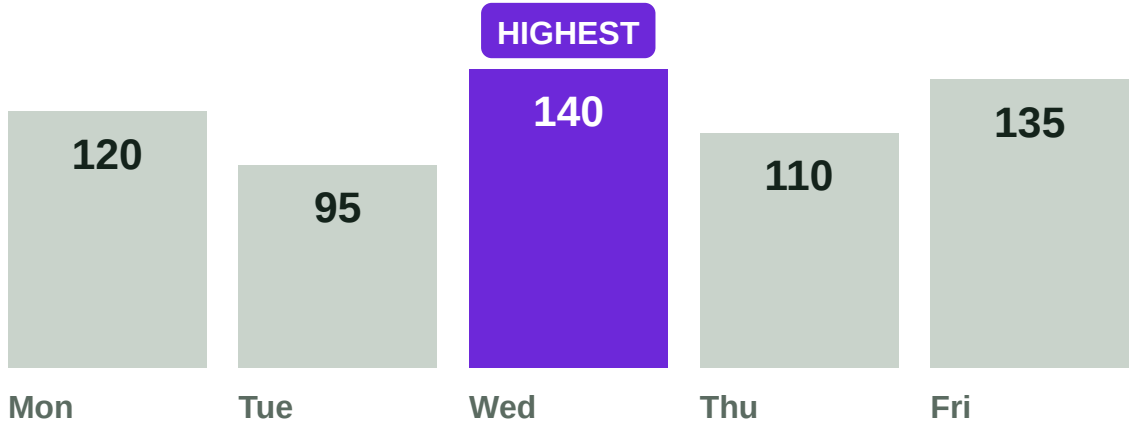
**WHAT IT DOES**

Finds the largest number in a range.

**EXAMPLE**

`=MAX(B2:B6)` → **140**

**THE TALLEST BAR IS THE MAX**




**Use it for:** the highest price, top mark or best day.

**Watch out:** MAX shows the value, not which day it was.

PRACTICE · WORK ALONE, THEN COMPARE WITH A PARTNER

# Practice: class markbook

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Student | English | Maths | Science | History | Total | Average | Lowest | Highest |
| 2 | Ali | 72 | 68 | 80 | 75 | | | | |
| 3 | Sara | 85 | 90 | 78 | 88 | | | | |
| 4 | Omar | 64 | 70 | 58 | 66 | | | | |
| 5 | Hina | 90 | 82 | 95 | 87 | | | | |
| 6 | Zain | 78 | 74 | 69 | 81 | | | | |

**Fill in:** `Total: SUM` `Average: AVERAGE` `Lowest: MIN` `Highest: MAX`

**Start with Ali:** his marks are in B2:E2.

**Exit ticket:** who has the highest total?